

Formulating Meaningful ML4H Research Questions

Emily Alsentzer

AHLI Summer Camp – Day 1



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The Taste Gap

Ira Glass, Creator
& Host of *This
American Life*
Podcast



Problem formulation is critical in the era of coding agents

Agent Laboratory: Using LLM Agents as Research Assistants

Samuel Schmidgall^{1, 2}, Yusheng Su¹, Ze Wang¹, Ximeng Sun¹, Jialian Wu¹, Xiaodong Yu¹, Jiang Moor³, Zicheng Liu¹ and Emad Barsoum¹

¹AMD, ²Johns Hopkins University, ³ETH Zurich

Article | [Open access](#) | Published: 25 March 2026

Towards end-to-end automation of AI research

[Chris Lu](#), [Cong Lu](#), [Robert Tjarko Lange](#), [Yutaro Yamada](#) ✉, [Shengran Hu](#), [Jakob Foerster](#), [David](#)

[Jeff Clune](#) ✉

[Nature](#) **651**, 914–919 (2026) | [Cite this article](#)

[nature](#) > [news](#) > article

NEWS | 19 May 2026

Teams of AI agents boost speed of research

Systems can generate hypotheses, interpret data and suggest ways to develop medicines.

[Home](#) > [Blog](#) >

Accelerating scientific breakthroughs with an AI co-scientist

February 19, 2025 · Juraj Gottweis, Google Fellow, and Vivek Natarajan, Research Lead

Outline for Today

- Defining research taste and impact
- Common pitfalls in problem formulation
- Questions to ask when beginning a new project
- Walkthrough of example ML4H projects

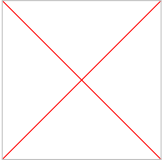
Research Taste

the tacit knowledge, intuitions, and judgments that guide a researcher's decisions throughout the research process

- Selecting important and tractable **problems**
- Designing informative **experiments**
- Recognizing **surprising results** worth pursuing
- Knowing **when to persist**, pivot, or quit

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What constitutes impactful research?

You must determine what kind of impact you want to have!



Belief

Changes what the field holds to be true

e.g. Shortcut learning research revealed failure mode in medical imaging models



Capability

Creates a tool, dataset, method, or result others can build on

e.g. Shared datasets / benchmarks or an open-source model



Agenda

Reframes what people work on by opening a subfield or closing a dead end

e.g. "Datasheets for Datasets" spurred research in dataset documentation



Practice

Changes what someone (e.g., clinician, regulator, etc.) does in practice

e.g. Obermeyer et al research on proxy labels shifted how practitioners considered bias in a deployed system.

Examples of Impactful Research

DOI:10.1145/345872

**Documentation to facilitate communication
between dataset creators and consumers.**

BY TIMNIT GEBRU, JAMIE MORGENSTERN,
BRIANA VECCHIONE, JENNIFER WORTMAN VAUGHAN,
HANNA WALLACH, HAL DAUMÉ III, AND KATE CRAWFORD

Datasheets for Datasets

Attention is not Explanation

Sarthak Jain

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Byron C. Wallace

Northeastern University

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Data Descriptor | [Open access](#) | Published: 24 May 2016

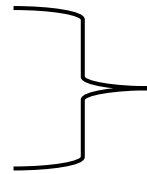
MIMIC-III, a freely accessible critical care database

[Alistair E.W. Johnson](#), [Tom J. Pollard](#) , [Lu Shen](#), [Li-wei H. Lehman](#), [Mengling Feng](#), [Mohammad Ghassemi](#), [Benjamin Moody](#), [Peter Szolovits](#), [Leo Anthony Celi](#) & [Roger G. Mark](#)

[Scientific Data](#) **3**, Article number: 160035 (2016) | [Cite this article](#)

377k Accesses | **7026** Citations | **177** Altmetric | [Metrics](#)

Healthcare cost as a proxy for health need produces a racially biased algorithm



The model learns to predict who can afford to go to get care, not who needs care

At the same level of algorithm-predicted risk, Blacks have significantly more illness burden than Whites

What constitutes impactful research?

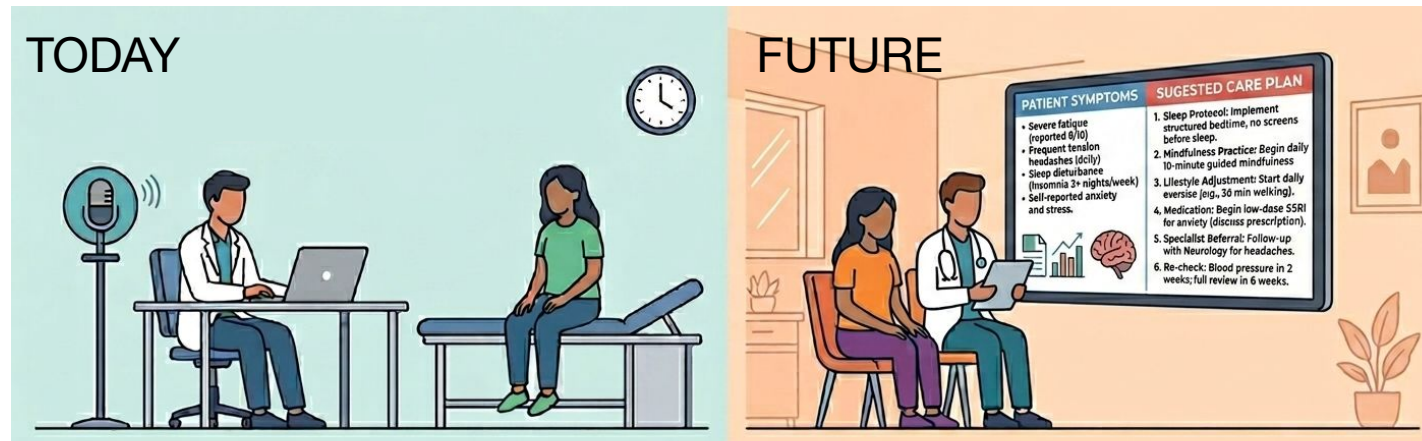
- **Reach:** how many people or decisions does it affect?
- **Durability:** does it contribute something that lasts the test of time?
- **Counterfactuality:** would this have appeared anyway within a year?
- **Actionability:** does a real decision change downstream?
- **Surprise:** does it challenge priors or confirm what everyone assumed?
- **Timing:** is it completed when it became possible or necessary?

What time horizon are you working towards?

Every project implicitly targets a future state of the world

Projects can optimize for

- Today's healthcare system vs. reimagined future healthcare system
- Available data vs. new data collection paradigms
- Near-term deployability vs. long-term transformation
- Publishable results vs. sustained impact



Let's Discuss!



- What kind of **impact** do you want your research to have?
- What **time horizon** does your work operate under?
- Which **ML4H papers** do you think have had the most impact? Why?

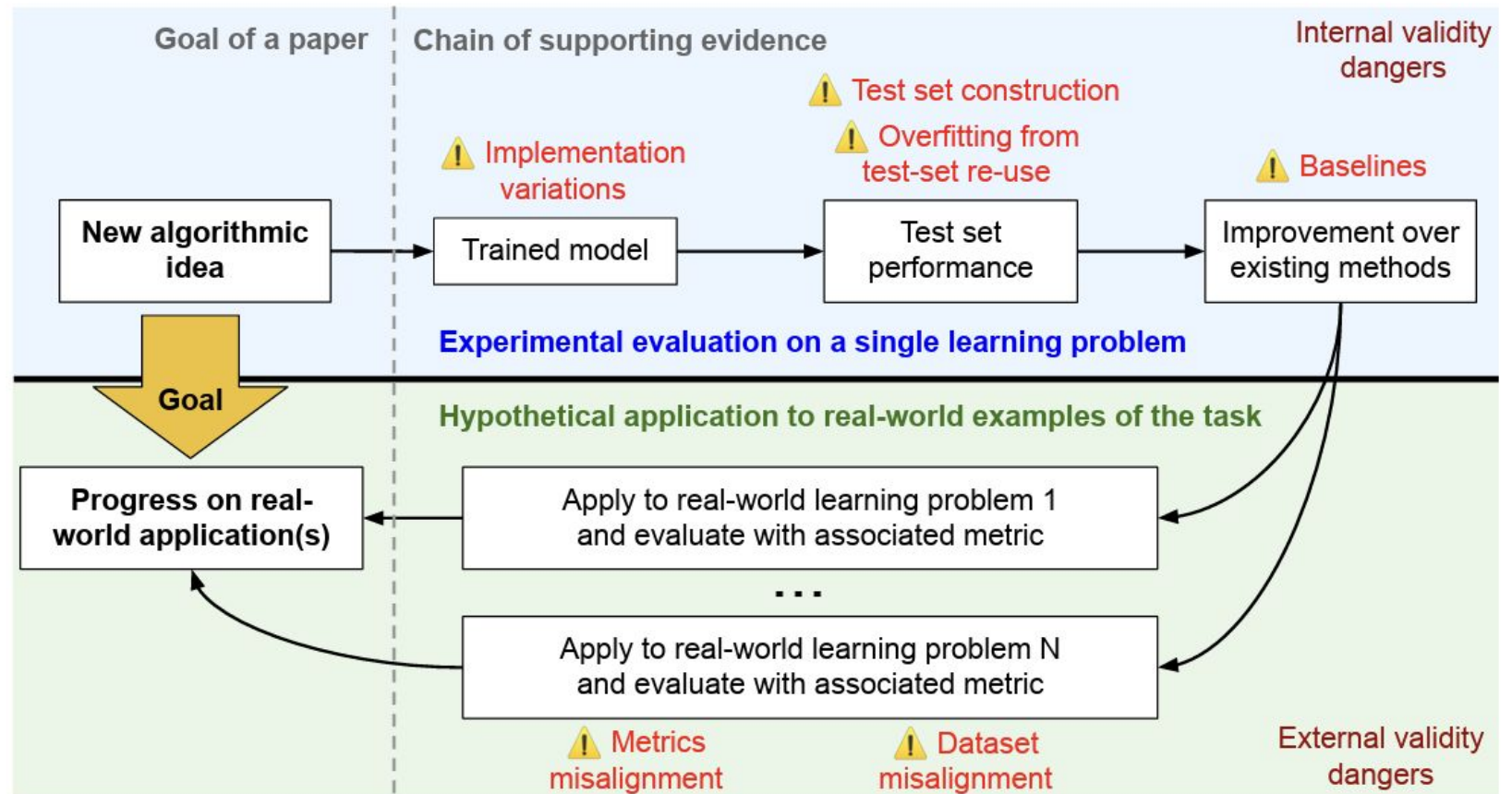
Common Pitfalls in Problem Formulation

Failures in Problem Formulation & Grounding

- **Key Question:** Is this a well-formed question, grounded in the actual problem, setting, and people it's for?
- Projects fail when they:
 - Tackle a problem that isn't the true bottleneck
 - Predict something clinicians already know
 - Predict outcomes that aren't linked to concrete actions
 - Optimize proxy targets rather than clinically meaningful goals
 - Produce tools that do not fit stakeholder workflows
 - Ignore stakeholder incentives and implementation constraints.
 - Leverage data unavailable at deployment or unrepresentative of real-world settings

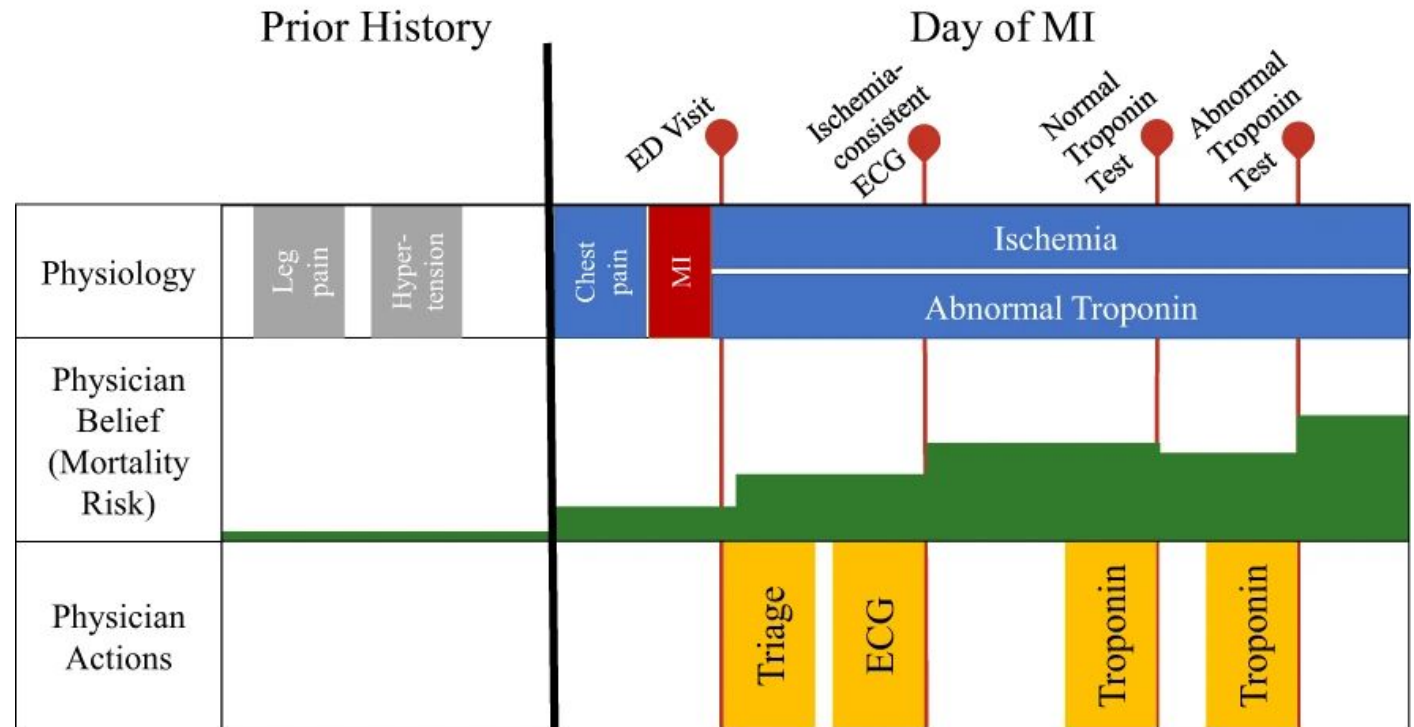
Failures in Problem Formulation & Grounding

Misalignment between benchmarks & real-world applications can lead to working on ill-posed problems



Failures in Problem Formulation & Grounding

Developing models that
“look over the
shoulders of clinicians”



ARTIFICIAL INTELLIGENCE

Hundreds of AI tools have been built to catch covid. None of them helped.

Some have been used in hospitals, despite not being properly tested. But the pandemic could help make medical AI better.

By Will Douglas Heaven

July 30, 2021

Why?

Model developers made incorrect assumptions about the data, which meant that the trained models did not work as claimed.

Driggs highlights the problem of what he calls Frankenstein data sets, which are spliced together from multiple sources and can contain duplicates. This means that some tools end up being tested on the same data they were trained on, making them appear more accurate than they are.

It also muddies the origin of certain data sets. This can mean that researchers miss important features that skew the training of their models. Many unwittingly used a data set that contained chest scans of children who did not have covid as their examples of what non-covid cases looked like. But as a result, the AIs learned to identify kids, not covid.

Driggs's group trained its own model using a data set that contained a mix of scans taken when patients were lying down and standing up. Because patients scanned while lying down were more likely to be seriously ill, the AI learned wrongly to predict serious covid risk from a person's position.

In yet other cases, some AIs were found to be picking up on the text font that certain hospitals used to label the scans. As a result, fonts from hospitals with more serious caseloads became predictors of covid risk.

Failures in Significance

- **Key Question:** If this research succeeds, what do we know that we didn't before and how much does that knowledge move the field?
- Projects can have limited significance when they:
 - Focus solely on building a model that beats other models on a benchmark
 - Show a delta too small to change anyone's conclusions
 - Report an improvement without explaining *why* it works
 - Over-optimize to a narrowly defined setting and fail to generalize
 - Demonstrate something others can't reproduce or extend
 - Improve on a baseline no one uses

Failures in Significance

Evaluation Metrics:

A. Technical replicability

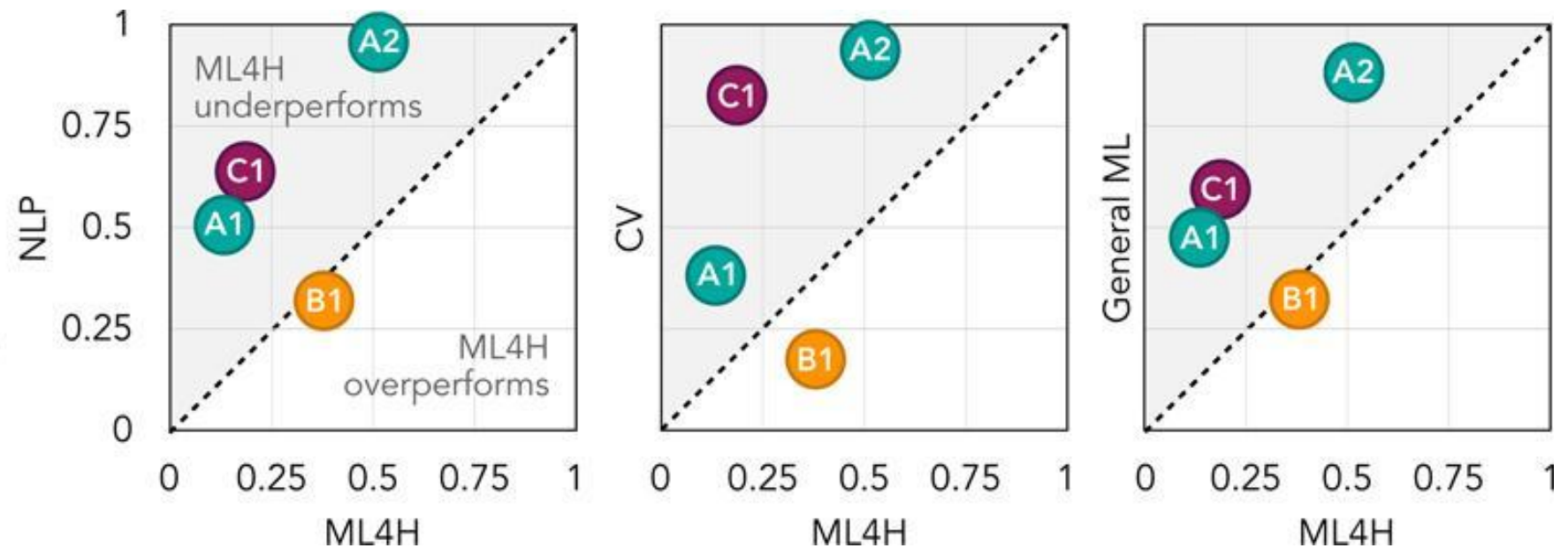
1. Code available
2. Public dataset

B. Statistical replicability

1. Variance reported

C. Conceptual replicability

1. Multiple datasets



Fraction of papers satisfying certain conditions by ML field

Failures in Timing

- **Key Question:** Why now? Will the outcome still matter when you're done?
- Projects fail when
 - A general advance will solve the problem as a byproduct before you finish
 - An external change (e.g., new data source, guideline, regulation) obviates the question while it's in progress
 - The contribution has a short shelf life. it won't hold or matter beyond a brief window
 - Someone else completes your project before you do

Avoiding Failures in Timing



"Skate to where the puck is going, not where it has been." - Wayne Gretzky

Failures in Feasibility

- **Key Question:** Can the project be completed now and can you do it?
- Projects fail when they:
 - Depend on data that doesn't exist, isn't accessible, or can't be obtained at the scale and quality needed
 - Require methods that aren't mature or available yet
 - Ignore why prior work failed
 - Don't align with your key advantages (data, knowledge, partnerships)
 - Have no viable minimum version

Let's Discuss!



- What kind of failure modes have your research projects been susceptible to?
- What's your unfair advantage—data, expertise, collaborators—and what questions does it uniquely position you to ask?

Questions to ask when
beginning a new project

Project Planning Questions

Problem: What's the problem? Why is it important? Why is it challenging?

Status Quo: What has been done previously to address the problem?

Gap: What is the gap in existing solutions / approaches? Why hasn't this gap been addressed yet?

Innovation: What novel idea, approach, or insight are we proposing? Why is this innovation expected to overcome existing limitations or challenges?

Impact: What broader impacts will our solution have if successful? Who will benefit from this work, and in what ways? How will this solution influence future research or practice?

Project Planning Questions

Methods: How specifically will we implement or test this innovation? What methods or techniques will be used, and why are they appropriate?

Data: What dataset / cohort will we use to implement or test the innovation?

Expected Contributions: What are the anticipated outcomes or deliverables of this work? How will these contributions advance the field?

Evidence/Experiments Needed: What experiments or evidence are required to validate our approach? What criteria will we use to determine if we have succeeded?

Risk & Mitigation: What potential challenges or limitations can we anticipate? What contingency plans or alternative strategies are available? How will we assess whether these mitigation strategies are effective?

Example ML4H Projects

Readmission Prediction

The problem

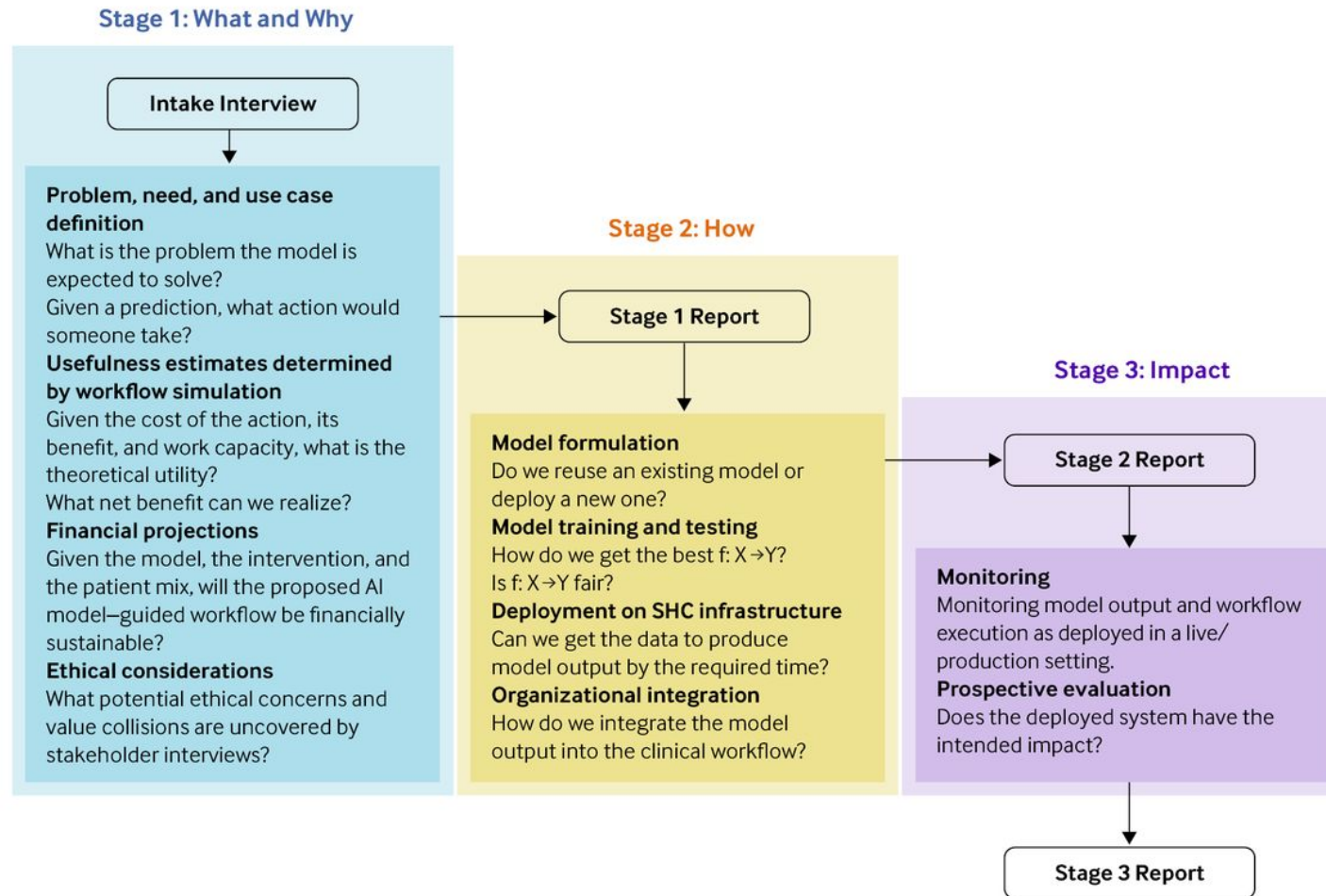
- Hospital A's readmission rates are rising. Readmissions are disruptive to patients & caregivers, costly to healthcare system, and put patients at additional risk of hospital acquired infections and complications
- Hospitals with high unplanned readmission rates face penalties for unplanned, preventable readmissions that occur soon after discharge. E.g., CMS reduces reimbursement for all Medicare patients if a facility has a higher-than-expected 30-day readmission rate.

Initial research question: Can we predict 30-day readmission for patients discharged from hospital A?

Challenge: Preventing readmissions requires understanding which patients can benefit from specific programs, not just which patients will be readmitted

Revised research question: Can we predict *modifiable* 30-day readmission for patients discharged from hospital A? Can we predict

At Stanford, AI models undergo Fair, Useful, and Reliable Model (FURM) Assessments

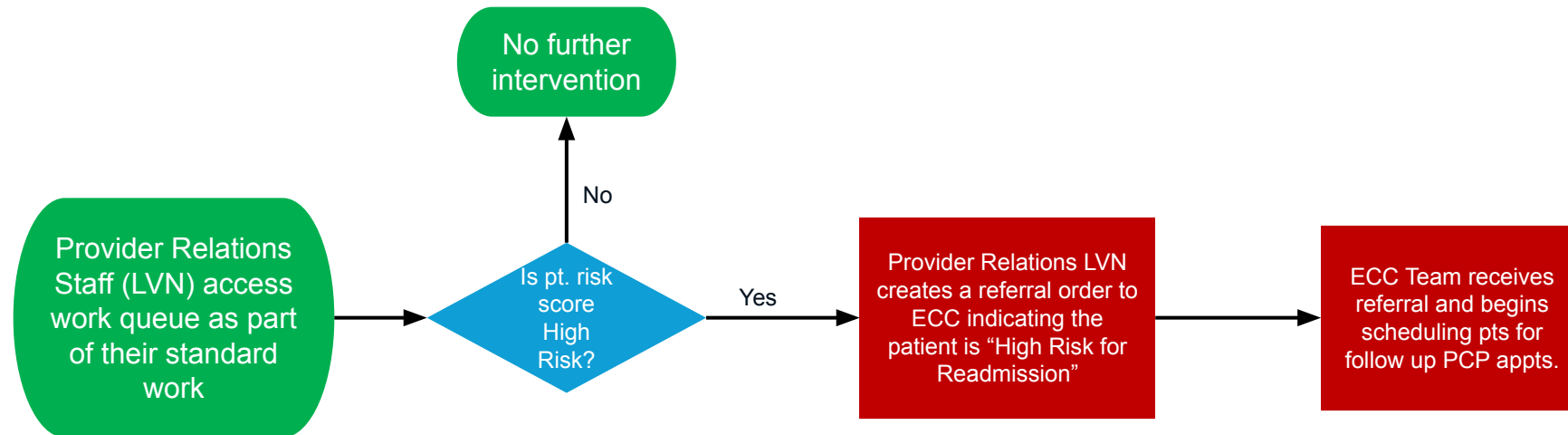


- 21 FURM assessments conducted since 2022
- 16/21 models were deployed
- Example: Epic's V2 readmission model

Key Stakeholders & Solution Developers

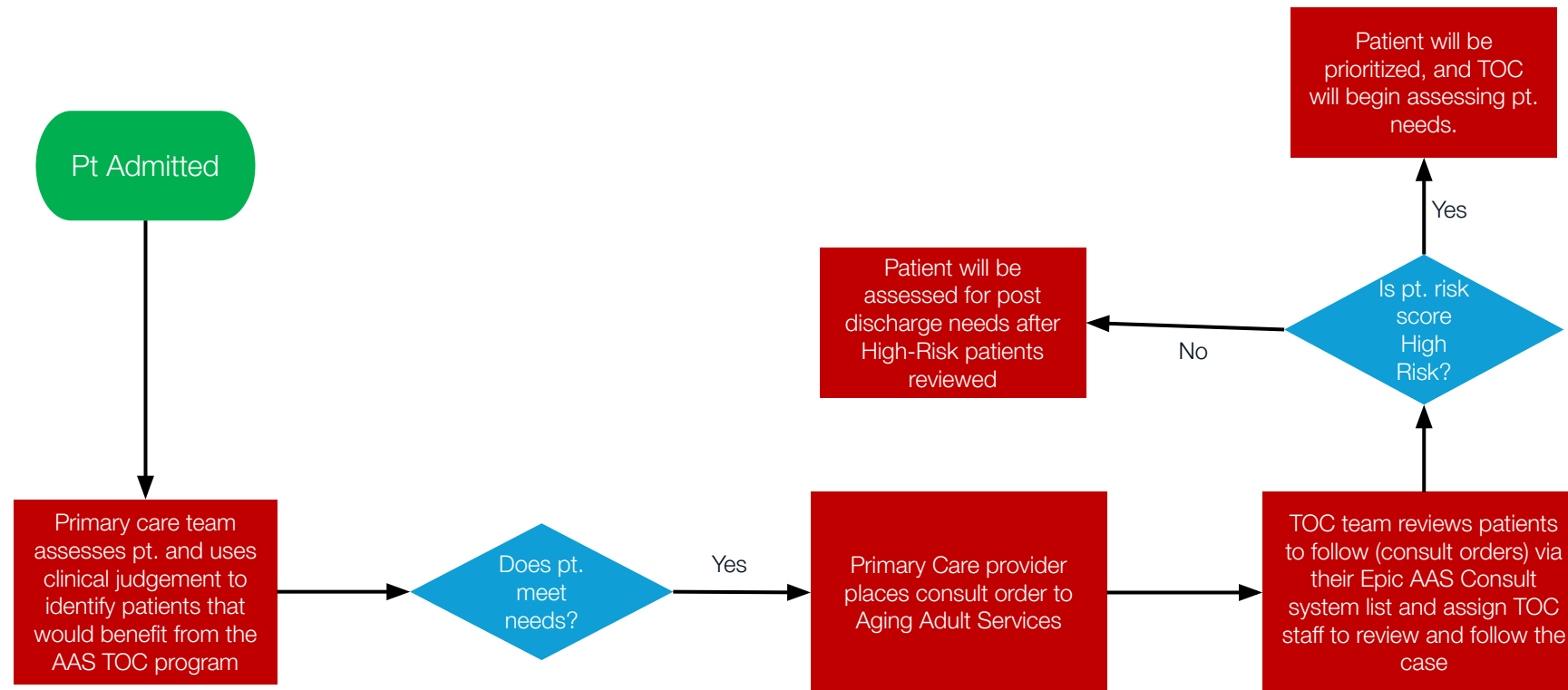
Role	Name
Sponsors	Readmission MGT Sponsors:
Operational Stakeholders	• ACQO • VP – Care Continuum • Director – Professional Practice • CNIO
Clinical Stakeholders	• Director of Transitional Care • Assistant Manager of Employer Based Clinics
Technical Stakeholders	• Director – TDS Inpatient Portfolio • Team Lead, IT Clinical Applications - Inpatient
Medical Informatics	• OB/GYN Inpatient & Ambulatory • Inpatient/OR
Nursing Informatics	Sr. Program Manager

Intended use of a model in a workflow



Enterprise Contract Center (ECC) Appointments Before Discharge Program
Only high-risk patients receive follow up appointments prior to discharge

Intended use of a model in a workflow



Adult Aging Services (AAS) Transition of Care Patient Referrals
Staff leverage the readmission score ONLY to prioritize the order in which they review patients referred to program

Intended use of a model in a workflow shapes evaluation

Option 1

Workflow

High risk patients receive 1:1 support from case manager

Metric

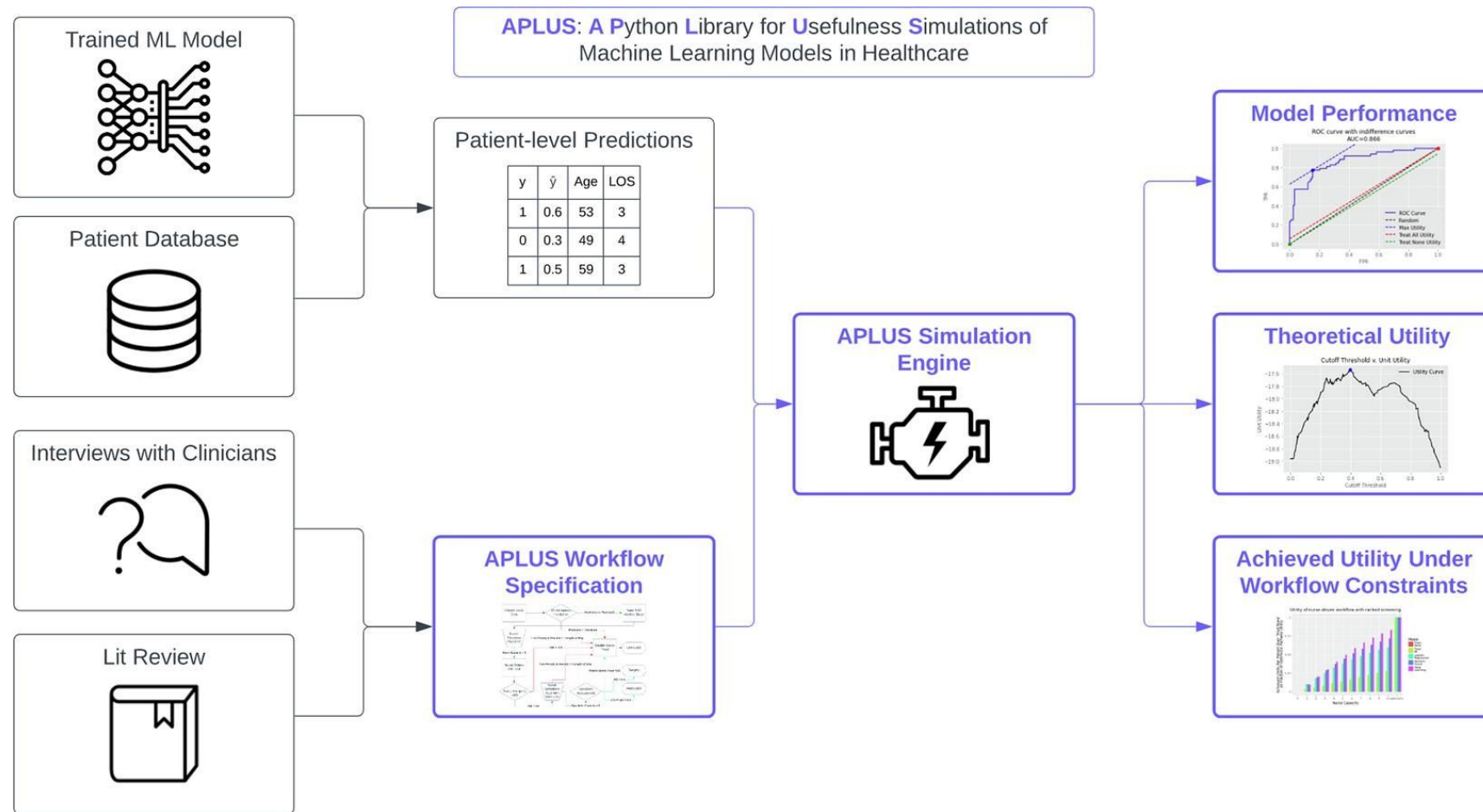
Precision at k
Where k is determined by capacity of case managers

Option 2

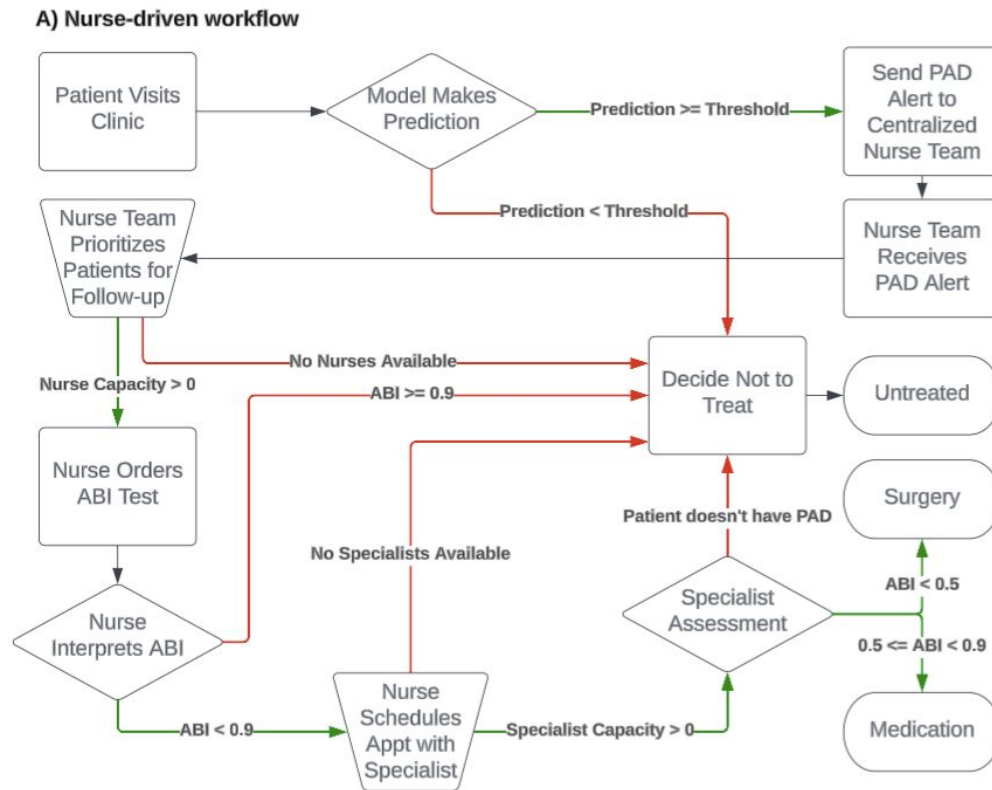
High risk patients are included in priority queue for follow up appointments

Recall

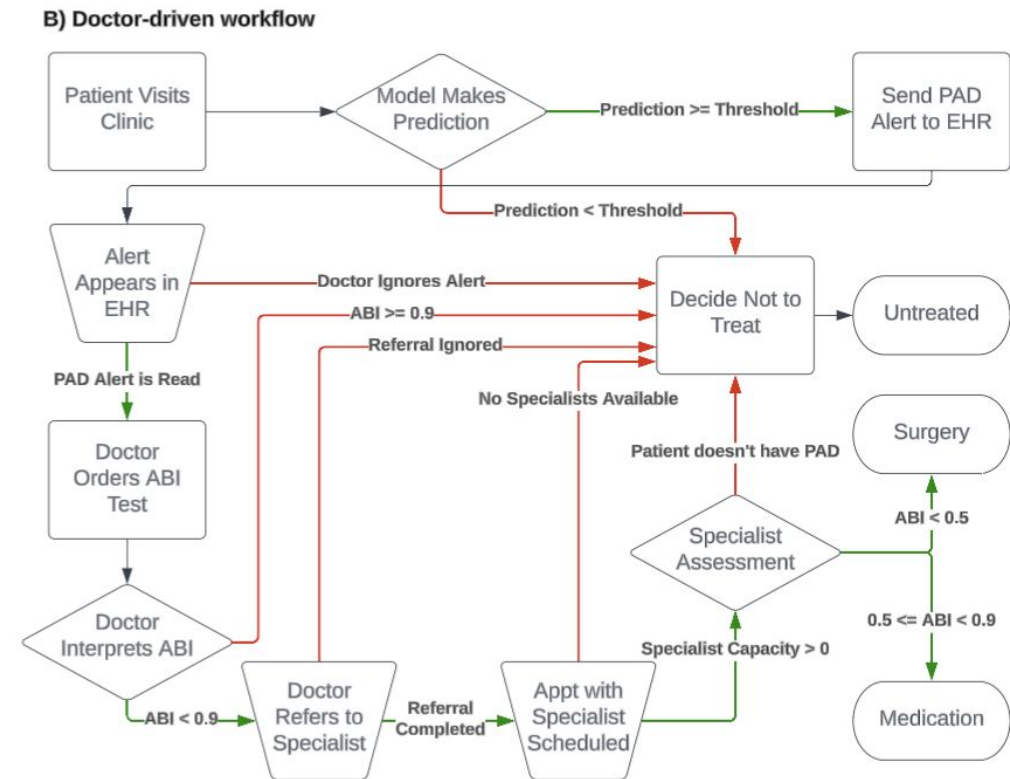
Assessing usefulness via workflow simulations



Assessing Nurse vs Doctor-driven Workflows for Peripheral Artery Disease (PAD) Screening



Centralized team of nurses reviewing the PAD model's predictions for all patients



PAD model's predictions appear as a real-time alert in a patient's EHR during their visit to the clinic

Generating Discharge Summaries

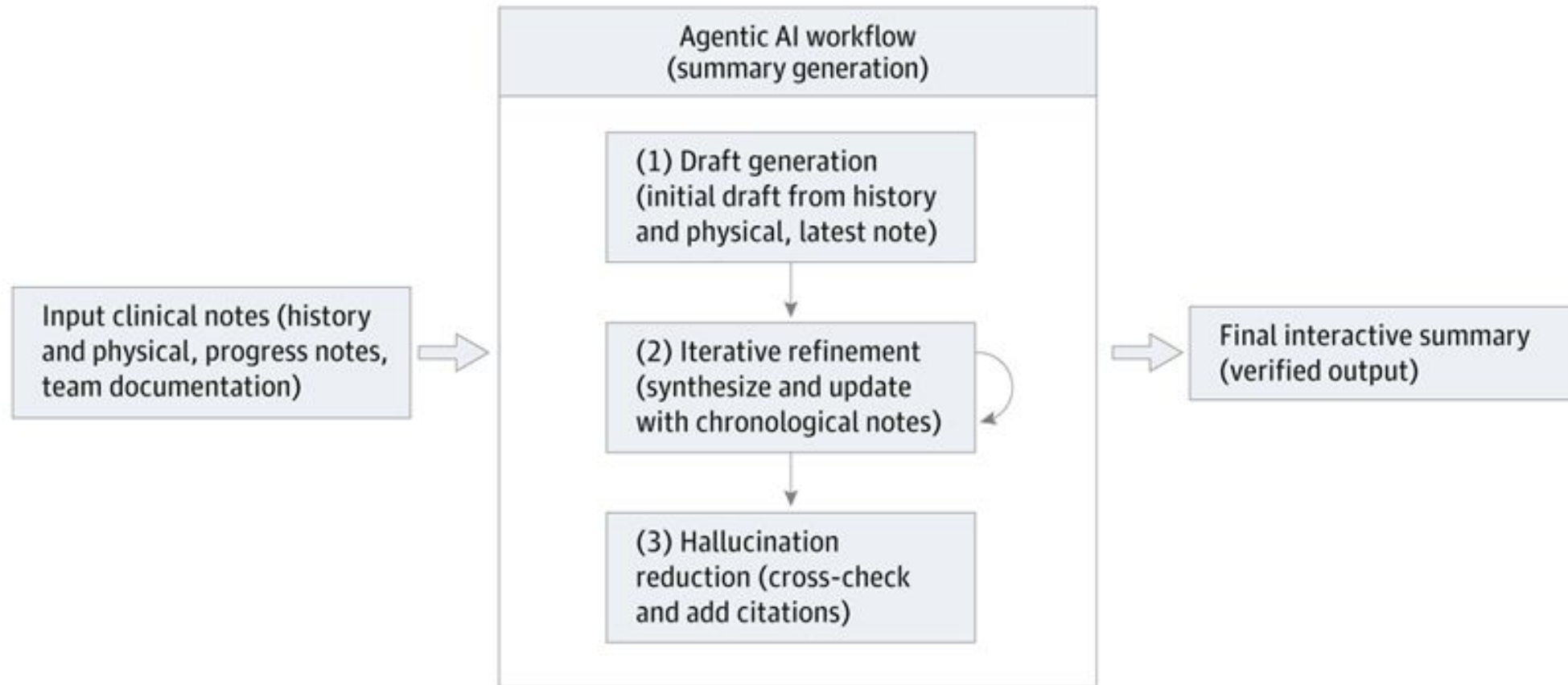
The problem: Discharge summaries summarize the events that happen to a patient during an inpatient stay. Generating discharge summaries is time consuming for clinicians and can delay discharge.

Initial research question: Can we use LLMs to generate the full discharge summary right before the patient is discharged?

Challenge: Most of the discharge summary is auto-generated from structured data! Clinicians draft the discharge summary throughout the patient's hospital stay

Revised research question: Can we use LLMs to generate the brief hospital course section of the discharge summary? Can we generate interim discharge summaries every day?

Generating Discharge Summaries



Be wary of incumbents if you are aiming for deployment

Artificial Intelligence (AI)

Roadmaps give you the big picture about key features that are already available and our broad vision for where we're headed. Many of our unreleased features are still being designed and do not yet have a planned release date. It's possible features might take a different approach than originally described here, might not be released for a while, or might be deprioritized.

Want to learn more? [Related Documents](#)

Ready for Broad Adoption	Pioneering	In Development and Future Plans
Staying Current (with Upgrade) ★ In Basket Art: Draft Responses to Patient Messages ★ Outpatient Insights ★ AI Text Assistant ★ Draft Hospital Course in Discharge Summary ● Actionable Follow-Ups for Radiology ★ Ambient Voice Documentation for Outpatient Notes ★ Patient-Facing Inpatient Instructions ★ Draft Answers for Prescription Prior Authorizations ★ Automatic Adenoma Detection Rate ★ Automatic Sessile Serrated Lesion Detection Rate ★ Extracted Sig Details ★ Discrete Data Extraction for Synoptic Forms ★ Transplant Evaluation Insights ★ Anesthesia Pre-Procedure Insights ★ Preprocedural Screening Insights ★ Dialysis Rounding Insights ★ Draft Invasive Procedure Narrative Staying Current (Generally Available) ★ Ambient Voice Recognized Ordering ★ Ambient Voice Documentation for Inpatient Notes ★ Ambient Voice Documentation for Problem-Oriented Charting ★ Inpatient Insights ★ In Basket Art: Draft Patient-Facing Result Comments	Pioneering (Includes Fit Considerations) ★ Ask Art ★ Ambient Voice Documentation for Nursing ★ Synopsis Insights ★ Draft Patient Instructions from Notes ★ Extract Social Drivers of Health from Notes ★ ED Facility Charge Edge Case Detection ★ Post-Procedure Diagnosis Suggestions ★ Cancer Staging Extraction from Notes ★ Draft Therapy Progress Note ★ Initial Patient Presentation Insights ★ Actionable Follow-Ups for Structural Heart Patients ★ Prior Study Comparisons ★ Nursing Assignment Agent Available for Early Feedback to Limited Sites ★ In Basket Art: Translations ★ Context-Driven In Basket Insights ★ Ambient Voice Documentation for OASIS Forms	In Development ★ Visit Agenda Agent ★ Ambient Voice Documentation for Tooth Charting ★ Ambient Voice Documentation for Orthopaedic Clinic Procedures ★ Ambient Voice Documentation for Endoscopic Findings ★ Ambient Voice Documentation for Skin Exams ★ Ambient Voice Documentation for Home Health and Hospice ★ Suggested Encounter Actions for Patient Messages ★ Colorectal Cancer Screening Follow-up Agent ★ Diagnosis Checker ★ Detect Sensitive Content in Notes ★ Extract Orders and Tasks from Notes ★ Draft Translations for AVS Instructions ★ Medication Usage Insights ★ Medication Verification Insights ★ Oncology Insights ★ Pregnancy Insights ★ Surgical Consult Insights ★ Surgical Case Length Estimation ★ Patient-Reported Outcome Insights ★ Group Session Insights ★ Extract Treatment History for Behavioral Health Notes ★ "How do I..." Search in Epic Assistant ★ Chronic Care Management Insights

Information Extraction from Clinical Notes

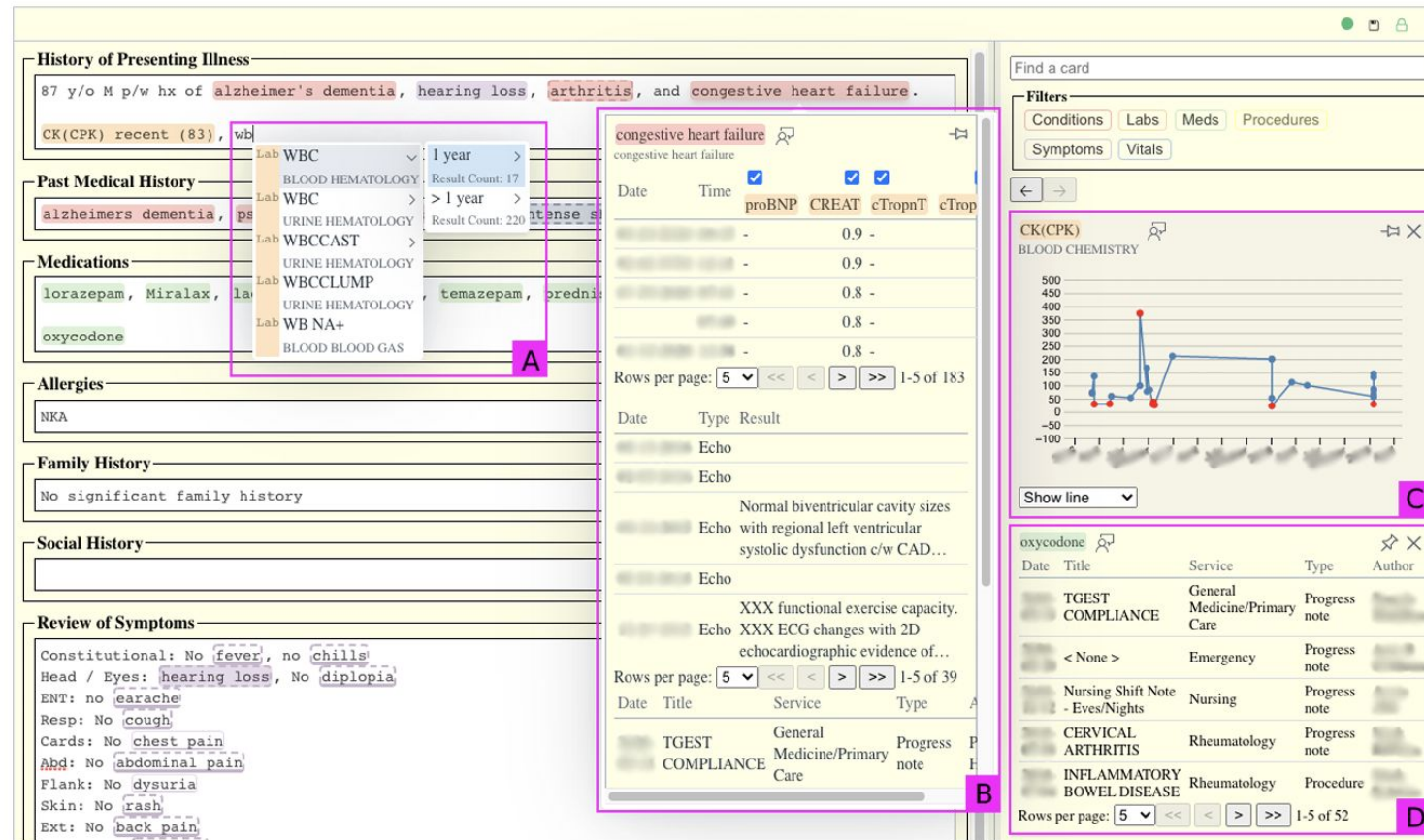
The problem: Substantial clinical information is only found in free-text longitudinal clinical notes. Extracting this information in a standardized way can enable downstream real world evidence research and clinical decision support applications.

Initial research question: Can we develop tools to extract information at scale from longitudinal clinical notes?

Challenge: Research question is designed as a band aid to current EHR workflows, rather than reimaging better workflows

Revised research question: Can we redesign EHR systems to automatically collect this structured data at data entry?

MedKnowts: Unified Documentation and Information Retrieval for Electronic Health Records



Rare Disease Diagnosis

The problem:

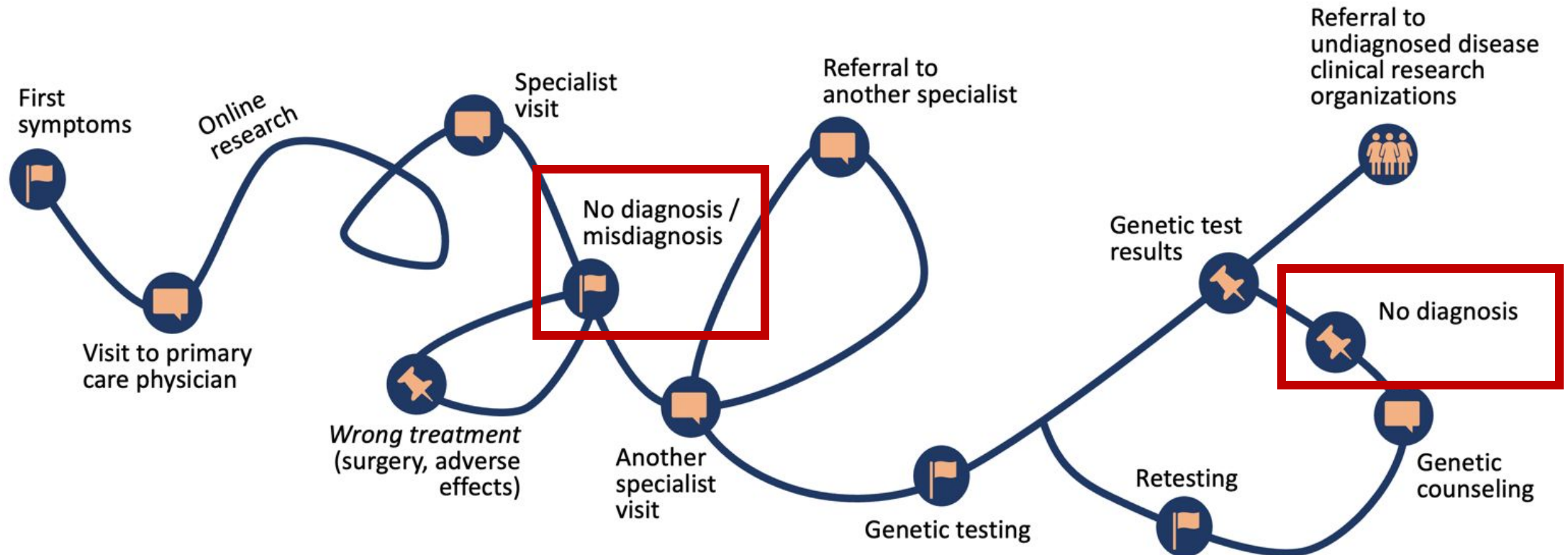
- There are approximately 300 million individuals with a rare disease
- Diagnosis is extremely challenging. There is substantial heterogeneity in clinical presentations, and clinicians have limited experience due to the low prevalence of each disease
- Identifying the molecular diagnosis requires integrating multiple sources of evidence to sort through thousands of candidate variants

Hypothesis:

- AI models can shorten the rare disease diagnostic odyssey

Rare Disease Diagnostic Odyssey

Where is the true bottleneck to diagnosis?



Average time to diagnosis: ~5-7 years (if ever)

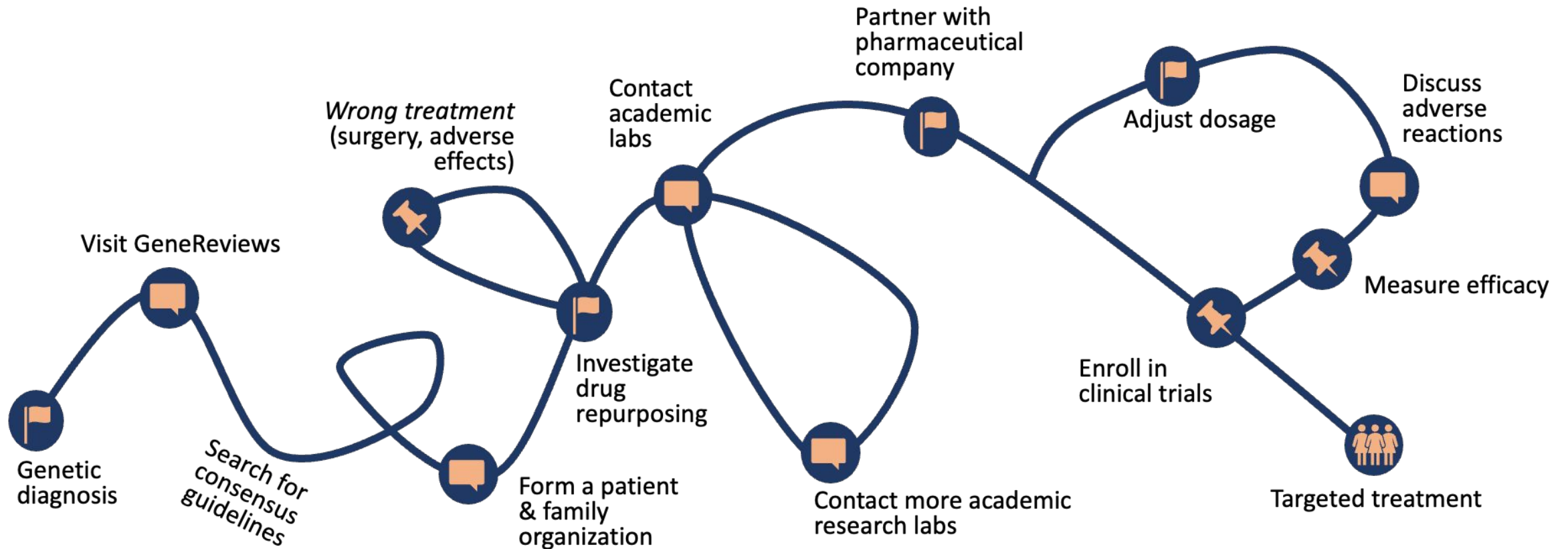
Number of specialists seen: >7

Diagnostic costs per patient: \$100,000+ (including misdiagnoses and unnecessary testing)

Emotional and psychological toll on patients and families

Adapted from The Marfan Foundation

Rare Disease Therapeutic Odyssey



Therapeutic options for rare diseases are limited.
Targeted treatments exist for ~5% of rare genetic diseases.*

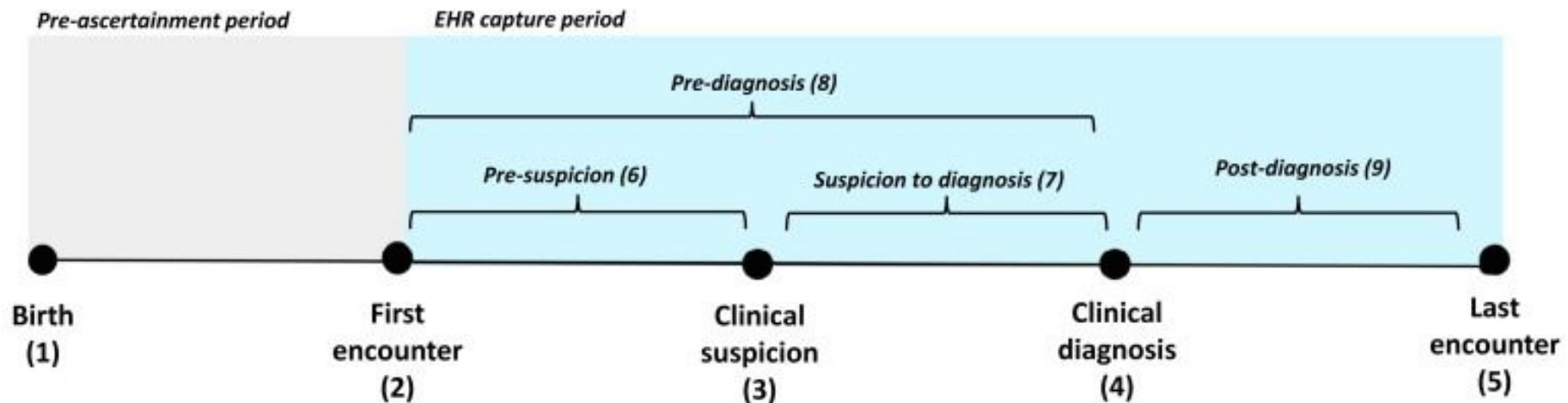
Make sure your model setup aligns with the problem formulation

Goal: Identify patients with undiagnosed rare diseases

Initial Setup: Train a model that takes longitudinal EHR data as input and outputs whether a patient has a particular rare disease d

Challenge: Model learns features from clinical suspicion period. Many rare diseases have the same downstream action.

Revised Setup: Train a model that inputs data **prior to diagnosis** and outputs whether a patient needs a **genetic testing workup**



Predicting Postpartum Hemorrhage

Align prediction task with meaningful clinical actions

● The Problem

PPH is the leading cause of maternal death worldwide, affecting over 14 million women annually and accounting for ~27% of maternal deaths globally. PPH rates are rising at Stanford.

Initial Research Question

- Can we predict which patients will experience PPH *one hour prior to delivery*?

● The Challenge

Most interventions only help *after* PPH onset. There's limited evidence for prophylactic drug administration, though more targeted type & screen offers some cost savings.

○ Revised Research Question

Can we predict PPH throughout the pregnancy course? Can we identify the factors driving rising rates?

Diagnostic & Therapeutic Decision Support

Align prediction task with meaningful clinical actions



The Problem

Demanding schedules and the difficulty of keeping pace with evolving medical evidence can lead clinicians to misdiagnose or mistreat patients.



Initial Research Question

Can LLMs support—or even perform—diagnostic and therapeutic decision-making? We'll evaluate this using readily available datasets.



The Challenge

Conclusions are highly sensitive to evaluation design such as, the choice of datasets, the clinicians involved, and how the clinician-in-the-loop setup is structured.



Revised Research Question

How can we design an evaluation that reflects how these models are used in clinical workflows?

Leveraging synthetic data for model training

The problem: Many clinical tasks have limited data available for model training. Synthetic data can improve the performance and robustness of models

Initial research question: Can we train GANs (or other generative models) on our dataset and use the synthetic data generated from the GAN to supplement our dataset for training?

Challenge: The GAN doesn't learn any new relationships that aren't already found in the initial data. Training on such data may provide regularization but no new knowledge

Revised research question: Can we inject external knowledge into the generative model beyond the original dataset to produce synthetic data?

Reinforcement Learning

How do your beliefs about the future state of the world shape what you focus on?

Hypothesis: Clinical data will remain siloed

Research: Focus on federated learning or model merging approaches

Hypothesis: General domain models will quickly be able to perform your clinical tasks without much domain specialization

Research: Focus on evaluation, deployment, human-AI interaction

Let's Discuss!



- What failure modes do you commonly see in the ML4H literature?
- What training do ML4H researchers need to avoid these common pitfalls?
- What bets are you making about the future in the work you are pursuing?

The Elephant in the Room: Incentives

The reward structure of ML/CS publishing optimizes for:

- Novelty over correctness
- Speed over depth
- Benchmarks over downstream impact
- Complexity over simplicity

Be cognizant of whether you are producing work that aligns with the impact you want to have

Formulating a Research Agenda

- During your career, you get the opportunity to place multiple bets on what the future will look like and how your work will move the field forward
- Cultivating a diverse research agenda with a mix of short-term and longer-term projects can be a practical approach to performing impactful work

Other Resources

- [“You and Your Research”](#) by Richard Hamming
- [“Research Taste Exercises”](#) by Chris Olah
- [“My Research Process: Understanding and Cultivating Research Taste”](#) by Neel Nanda

Questions?

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Appendix

Diagnostic & therapeutic decision support

The problem: Clinicians sometimes misdiagnose or mistreat patients due to demanding schedules and the challenges of keeping up-to-date with evidence in medicine.

Initial research question: Can we leverage LLMs to support (or perform) diagnostic and therapeutic decision making? Let's use readily available datasets for evaluation.

Challenge: The datasets and clinicians used for evaluation as well as the clinician-in-the-loop setup can drastically impact conclusions

Revised research question: Let's design an evaluation setting that reflects real-world use of these models in clinical workflows.

Predicting Postpartum Hemorrhage (PPH)

The problem: Postpartum hemorrhage is a leading cause of maternal morbidity and mortality. Postpartum hemorrhage rates are rising at Stanford.

Initial research question: Can we predict patients who will experience PPH one hour prior to delivery?

Challenge: Most inventions for PPH only provide benefit after PPH onset. There's limited evidence of utility of prophylactic drug administration. There's some cost savings of more targeted type & screen

Revised research question: Can we predict patients who will experience PPH throughout their pregnancy course? Can we identify factors that lead to increased rates of PPH?